

FILTER DESIGN TEST CASE

Duplexer Module - n77 (3.3-4.2 GHz 5G NR) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting n77 (3.3-4.2 GHz 5G NR) for Satellite L-Band Filter. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 4003.5 MHz and bandwidth of 23.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: n77 (3.3-4.2 GHz 5G NR)
- Application: Satellite L-Band Filter
- Substrate: 36° YX LiTaO3
- Package: LGA (Land Grid Array)
- Design Tool: Keysight ADS 2024
- Design Center: Andover HQ Lab

This specification applies to all variants of the Duplexer Module design targeting n77 (3.3-4.2 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4003.5 MHz
Bandwidth (3 dB):	23.1 MHz
Insertion Loss Target:	<= 1.1 dB
Return Loss Target:	>= 10.0 dB
Out-of-Band Rejection:	>= 47.0 dB
Isolation (Duplexer):	>= 40.0 dB
Q Factor:	11134
Temperature Coefficient:	-18.7 ppm/°C
Die Size:	1.6 x 2.3 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the Duplexer Module template project.
2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
3. Set design targets: center freq 4003.5 MHz, BW 23.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (LGA (Land Grid Array)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX LiTaO3).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 12010 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.1 dB
- Return loss in passband shall be >= 10.0 dB
- Out-of-band rejection at +/- 35 MHz offset >= 47.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 9.4 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 88%

6. TEST RESULTS

Measurement Date: 2025-05-05
Devices Tested: 57
Insertion Loss (meas): 0.85 dB
Return Loss (meas): 12.2 dB
Rejection (meas): 50.0 dB
Isolation (meas): 44.7 dB
Q Factor (meas): 11134
Group Delay Variation: 2.0 ns
Power Handling: +31 dBm
Die Yield: 81.6%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated excellent performance for n77 (3.3-4.2 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author: Dr. Ahmed Al-Farsi Date: 2025-05-05
Reviewer: Dr. Anya Krishnamurthy Date: 2025-05-08
Approver: Dr. Yuko Ishida Date: 2025-05-14

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